

EcoTrack AI: Carbon Footprint Awareness and Reduction Platform

Abstract

Climate change is one of the most pressing global challenges of the 21st century, with carbon emissions being a major contributor to environmental degradation. Many individuals unknowingly contribute to greenhouse gas emissions through transportation, electricity consumption, dietary habits, and daily lifestyle choices. However, there is often a lack of awareness and personalized guidance to help people understand and reduce their environmental impact.

EcoTrack AI is an intelligent Carbon Footprint Awareness Platform designed to educate users, measure their carbon emissions, provide AI-powered sustainability recommendations, and encourage environmentally responsible behavior. The platform combines carbon footprint calculation, AI-driven insights, sustainability education, progress tracking, and digital pledge certification into a single user-friendly web application.

By leveraging Google's Gemini AI, Firebase Cloud Services, and modern web technologies, EcoTrack AI empowers users to make informed decisions and adopt sustainable habits that contribute to a greener future.

Problem Statement

Individuals contribute significantly to carbon emissions through transportation, electricity usage, food consumption, and waste generation. Despite growing awareness of climate change, many people lack accessible tools that help them understand their personal carbon footprint and provide actionable recommendations for reducing their environmental impact.

Existing solutions often focus only on carbon calculation and fail to provide personalized guidance, educational resources, goal tracking, and user engagement mechanisms that encourage long-term behavioral change.

Therefore, there is a need for a comprehensive platform that enables users to:

- ★ Understand their carbon footprint.
- ★ Track carbon emissions over time.
- ★ Receive personalized sustainability recommendations.
- ★ Learn environmentally responsible practices.
- ★ Commit to sustainable actions through measurable goals.

Proposed Solution

EcoTrack AI addresses this challenge by providing an integrated platform that combines carbon footprint measurement, artificial intelligence, educational content, sustainability goal tracking, and community engagement.

Users can input their transportation habits, electricity consumption, and dietary preferences to calculate their estimated carbon emissions. Based on this data, the platform generates personalized sustainability recommendations using Google's Gemini AI model.

The platform also includes an awareness module that educates users about climate change, carbon emissions, and practical ways to reduce their environmental impact. Additionally, users can set carbon reduction goals, monitor progress through interactive dashboards, and take a sustainability pledge that generates a personalized digital certificate.

Objectives

The primary objectives of EcoTrack AI are:

- Increase awareness about carbon emissions and climate change.
- Help users measure their carbon footprint accurately.
- Provide personalized recommendations using Artificial Intelligence.
- Encourage sustainable lifestyle changes.
- Enable users to track and reduce their carbon emissions.
- Promote environmental responsibility through digital engagement and pledge initiatives.

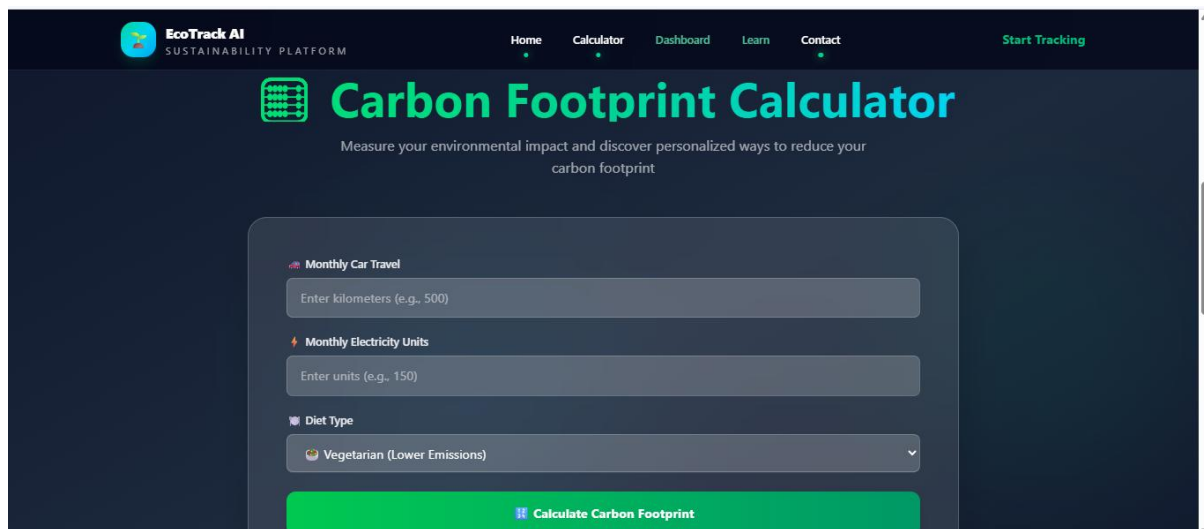
Key Features

1. Carbon Footprint Calculator

The platform calculates estimated carbon emissions based on:

- Monthly vehicle travel distance
- Household electricity consumption
- Dietary habits

Users receive an overall carbon footprint score represented in kilograms of CO₂ emissions.

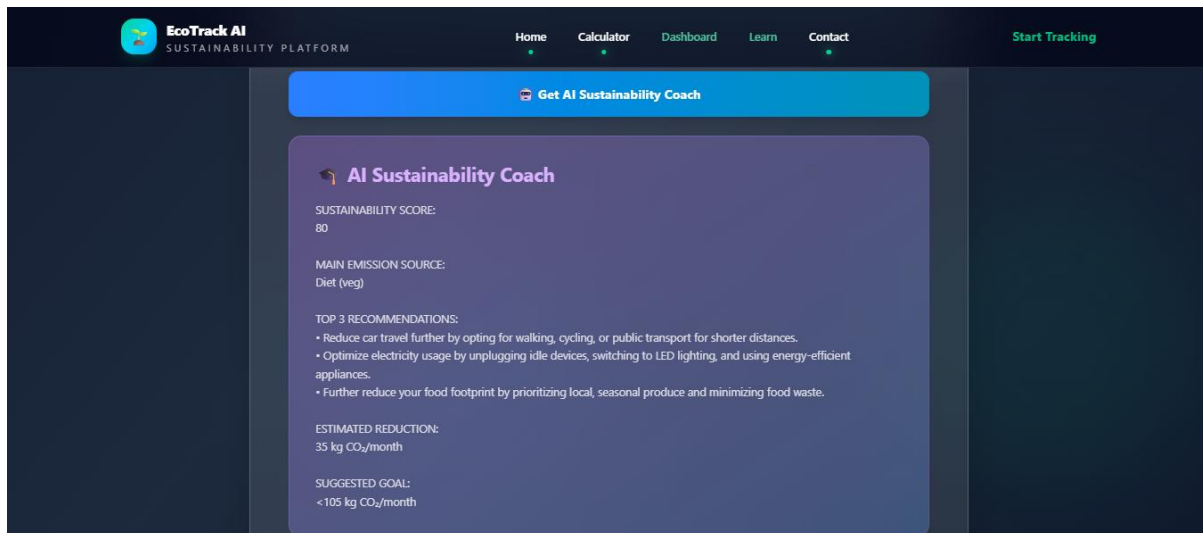
The screenshot shows the 'Carbon Footprint Calculator' interface on the EcoTrack AI Sustainability Platform. The header includes the logo, navigation links (Home, Calculator, Dashboard, Learn, Contact), and a 'Start Tracking' button. The main heading is 'Carbon Footprint Calculator' with a subtitle: 'Measure your environmental impact and discover personalized ways to reduce your carbon footprint'. The form contains three input sections: 'Monthly Car Travel' with a text field 'Enter kilometers (e.g., 500)', 'Monthly Electricity Units' with a text field 'Enter units (e.g., 150)', and 'Diet Type' with a dropdown menu currently set to 'Vegetarian (Lower Emissions)'. A green button at the bottom is labeled 'Calculate Carbon Footprint'.

2. AI Sustainability Coach

EcoTrack AI integrates Google Gemini AI to provide personalized sustainability insights.

The AI Coach:

- Analyzes user carbon footprint data.
- Identifies major emission sources.
- Suggests practical emission reduction strategies.
- Recommends achievable sustainability goals.
- Estimates potential carbon reduction.



3. Sustainability Dashboard

The dashboard provides visual insights into user emissions.

Features include:

- Total carbon footprint.
- Emission category breakdown.
- Interactive bar charts.
- Pie chart visualizations.
- Latest assessment summary.

This helps users better understand their environmental impact through data-driven visualization.

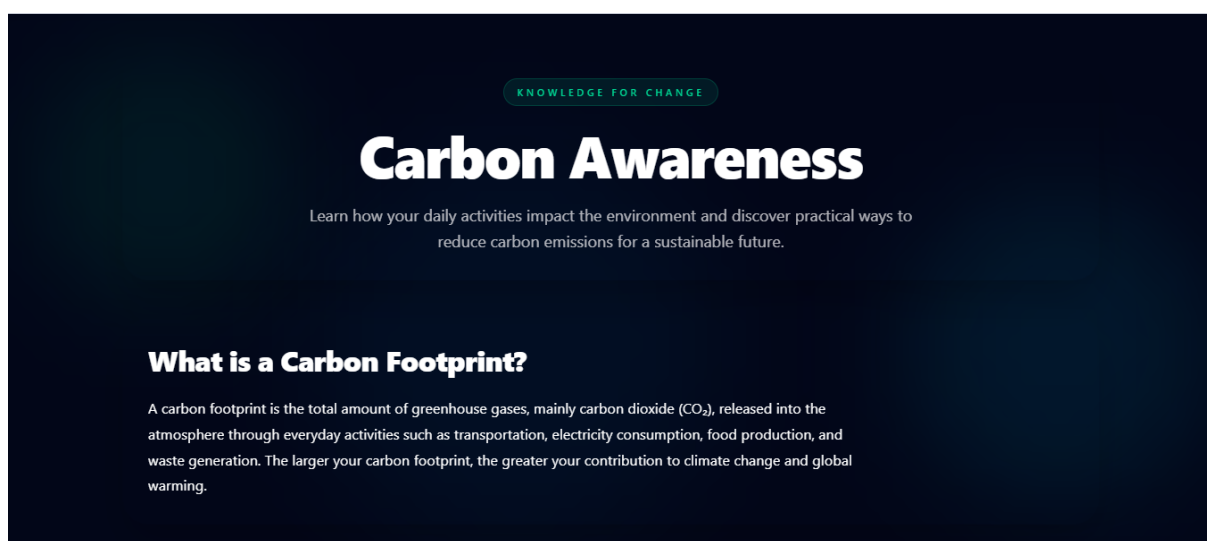


4. Awareness and Education Module

The awareness section educates users about:

- Carbon footprint fundamentals.
- Climate change impacts.
- Sources of carbon emissions.
- Sustainable living practices.
- Environmental conservation strategies.

This module promotes long-term environmental awareness and behavioral change.



5. Carbon Reduction Goal Tracker

Users can define sustainability goals and track progress toward achieving them.

The system allows users to:

- Set emission reduction targets.
- Monitor progress.
- Visualize achievement percentages.
- Encourage accountability and continuous improvement.

6. Green Pledge System

Users can take a sustainability pledge committing to environmentally responsible practices.

Features include:

- Digital pledge registration.

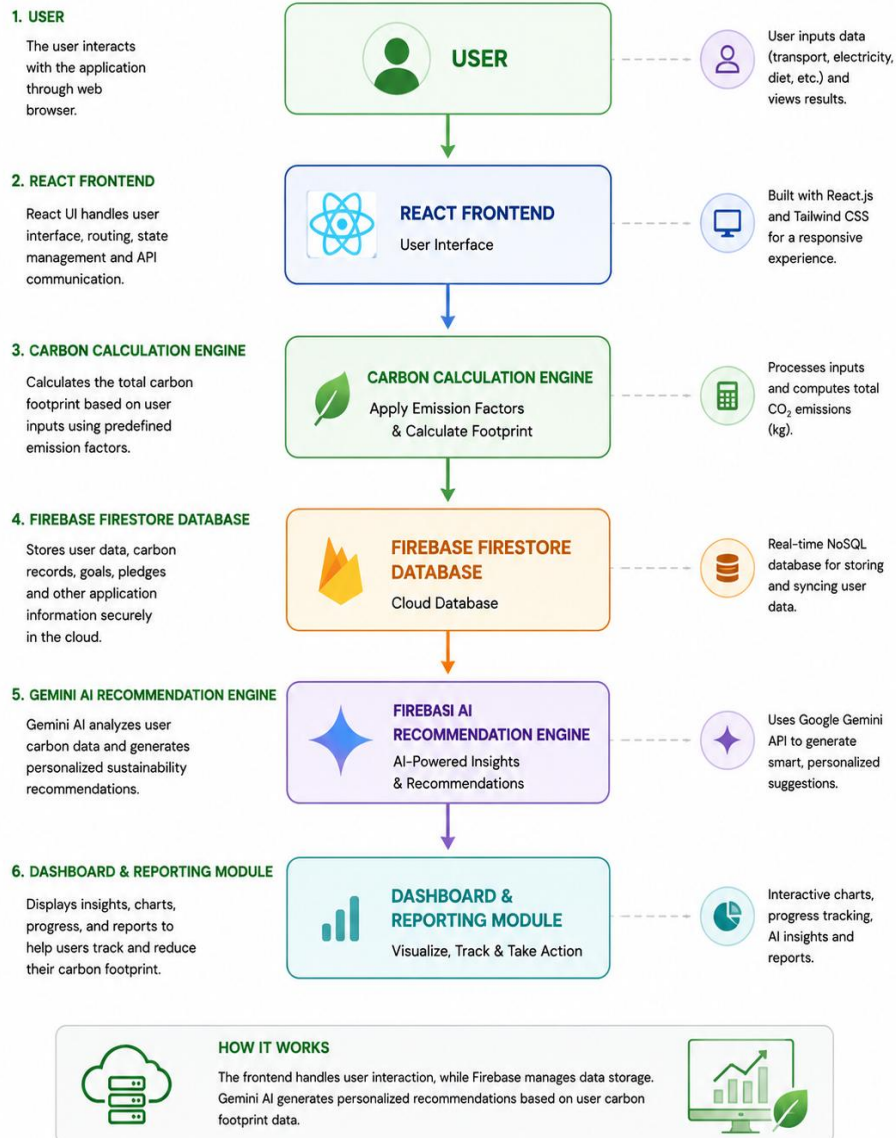
- Personalized certificate generation.
- QR-enabled verification capability.
- Downloadable PDF certificates.

This feature increases user engagement and promotes environmental commitment.

System Architecture

EcoTrack AI – System Architecture

Client-Cloud Architecture for Carbon Footprint Awareness Platform



Technology Stack

Frontend:

- React.js
- Tailwind CSS
- Recharts

Backend Services:

- Firebase Firestore

Artificial Intelligence:

- Google Gemini API

Database:

- Firebase Firestore

PDF Generation:

- jsPDF
- Html2canvas

Deployment:

- Vercel

Version Control:

- GitHub

Expected Impact

EcoTrack AI aims to create a positive environmental impact by:

- ★ Increasing climate awareness.
- ★ Encouraging sustainable behavior.
- ★ Reducing individual carbon emissions.
- ★ Promoting responsible consumption.
- ★ Supporting global sustainability goals.

The platform transforms environmental awareness into actionable insights, helping users make informed decisions that contribute to a greener and more sustainable future.

Conclusion

EcoTrack AI is an AI-powered Carbon Footprint Awareness Platform that enables individuals to understand, monitor, and reduce their environmental impact. By combining carbon footprint calculation, AI-driven sustainability coaching, educational content, goal tracking, and digital pledge certification, the platform provides a comprehensive solution for promoting sustainable living.

The project demonstrates how Artificial Intelligence and modern web technologies can be leveraged to address environmental challenges and encourage meaningful climate action at an individual level.